

Session 05 – AI Agents

Dr Ivan Olier

I.Olier@ljmu.ac.uk

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Large Language Models

- Trained on vast text data to process and generate human language.
- Based on deep learning (e.g., transformers like GPT, BERT, LLaMA).
- Key Capabilities:
 - Text generation (e.g., chatbots, content creation)
 - Summarisation & translation
 - Question answering & information retrieval
 - Text classification (e.g., sentiment analysis)
 - Code generation & automation

 LLaMA  Claude

ChatGPT

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Model Providers and Inference Platforms

Groq

Fast hosted inference for supported models

OpenAI API

Hosted models, tools and agent workflows

Google Gemini API

Gemini models and Google AI Studio

Anthropic Claude API

Claude models with tool use and structured workflows

Hugging Face

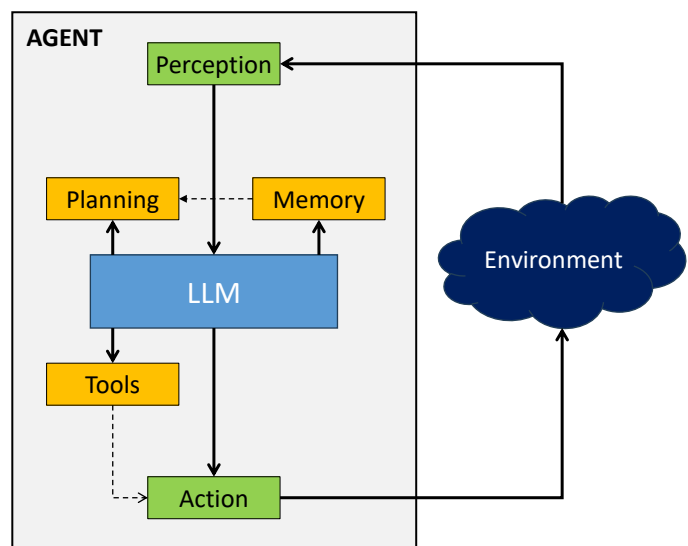
Model hub and inference provider ecosystem

Ollama

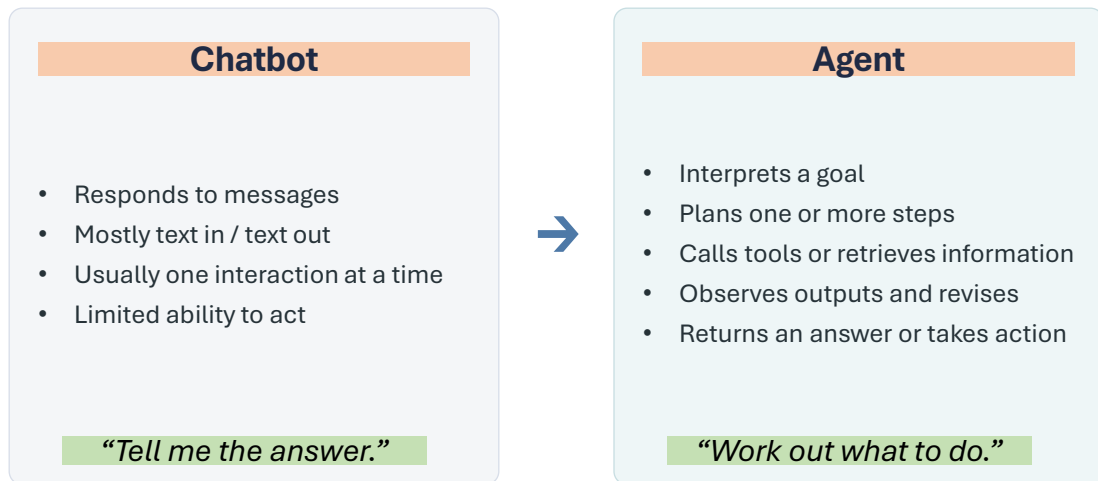
Run open-weight models locally

LLM-Powered Agents

- Leverage Large Language Models (LLMs) for reasoning, decision-making, and task automation.
- Key Features:
 - Natural Language Understanding (NLU) – Interprets and generates human-like text.
 - Reasoning & Planning – Can break down complex tasks and adapt strategies.
 - Autonomy – Executes tasks independently with minimal human intervention.
 - Memory & Context Awareness – Retains and uses past interactions for better responses.



Agent vs Chatbot

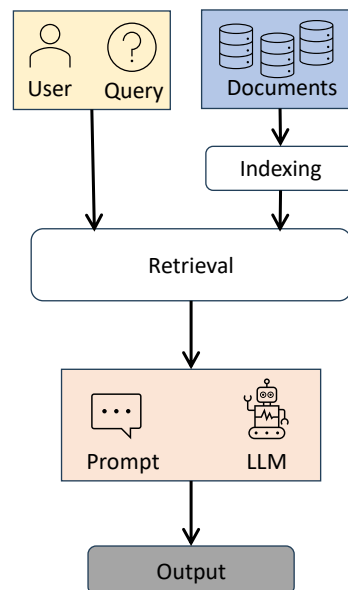


AI Agent Frameworks

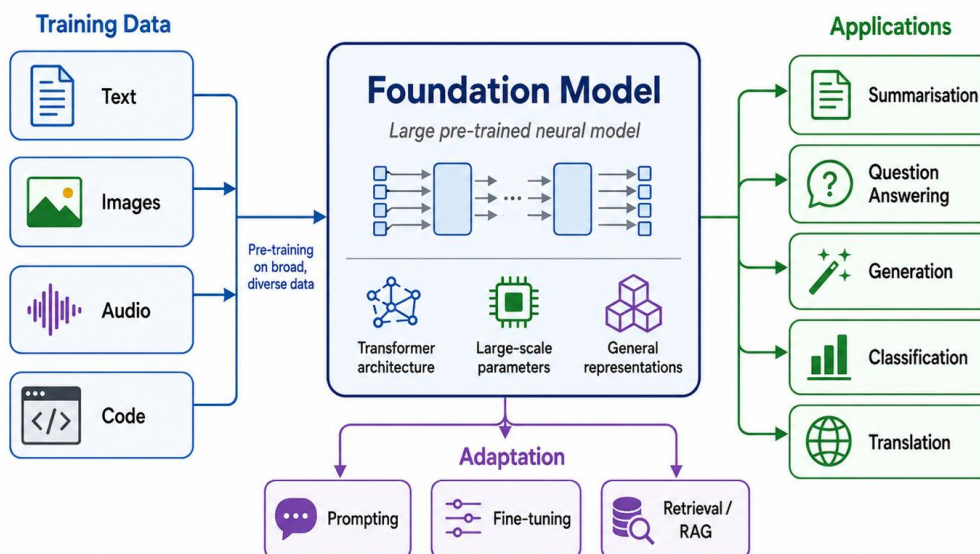
- Common AI Agent frameworks:
 - **LangChain, OpenAI Agents SDK, Google ADK, Microsoft AutoGen, CrewAI, ...**
- How they typically work:
 - The framework sends the LLM a description of every available tool (name, purpose, inputs, outputs).
 - The LLM decides which tool is appropriate based on the user's request.
 - It returns a **structured function call** (often JSON).
 - The framework parses that structured output, invokes the corresponding Python function, collects the result, and optionally sends the result back to the LLM for further reasoning.
 - The process can repeat multiple times until the LLM indicates that it has enough information to produce the final answer.

Retrieval-Augmented Generation (RAG)

- Combines a retrieval system with an LLM.
- Before generating text, the model retrieves relevant documents (from a database, API, etc.) to ground its output in up-to-date or domain-specific knowledge.
- Reduces hallucinations and extends the model's effective knowledge base beyond its training data.
- Example: Using RAG for medical Q&A, where the LLM fetches updated clinical guidelines before generating a response.



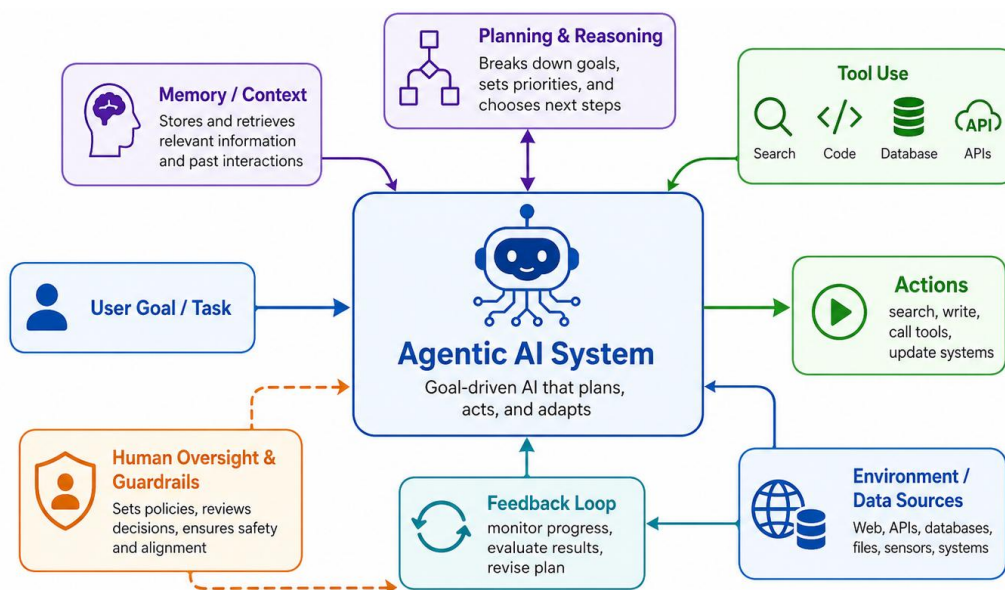
Foundation Models



Foundation Models

- **Foundation models** are large AI models trained on broad, diverse datasets so they can be adapted to many downstream tasks, rather than being built for one narrow purpose.
- They learn **general-purpose representations** of language, images, audio, code, or multimodal data, enabling tasks such as classification, generation, summarisation, translation, reasoning, and question answering.
- Their main components include a **large neural architecture** often transformer-based, massive training data, self-supervised or weakly supervised pre-training, and task adaptation through prompting, fine-tuning, or retrieval-augmented generation.
- They are important because they provide a reusable AI capability layer, reducing the need to train models from scratch and enabling faster development of intelligent applications across sectors.

Agentic AI



Agentic AI

- **Agentic AI** refers to AI systems that can pursue goals by planning, making decisions, taking actions, and adapting their behaviour with limited human intervention.
- Unlike standard generative AI, which mainly responds to prompts, agentic AI can **break down tasks**, choose tools, call external systems, monitor progress, and revise its strategy.
- Core components include a **large language model**, memory, planning/reasoning mechanisms, tool access, feedback loops, and guardrails for safety, reliability, and human oversight.
- Its importance lies in enabling more autonomous workflows, such as research assistants, coding agents, clinical decision-support assistants, customer-service agents, and operational process automation.
- Agentic AI builds on decades of research in intelligent agents and multi-agent systems, but the term became widely used in the generative AI era, especially from 2023 onwards, to describe LLM-based systems that can plan, use tools, act, and adapt with limited human intervention.

Agentic AI platforms

- **Provider-native agent APIs:**
 - OpenAI Agents SDK / Responses API, Anthropic Claude API, Google Gemini / Vertex AI Agent Builder, Amazon Bedrock Agents, and Microsoft Azure AI Foundry.
- **Workflow and orchestration frameworks:**
 - LangGraph, CrewAI, LlamaIndex, Semantic Kernel, and AutoGen help coordinate planning, memory, tool use, retrieval, and multi-agent workflows.
- **What they provide:**
 - access to LLMs plus tools, APIs, files, databases, web/search, code execution, memory, retrieval-augmented generation, monitoring, and guardrails.
- **How to choose:**
 - use provider-native APIs for quick deployment, orchestration frameworks for customised workflows, and enterprise platforms when governance, security, scalability, and monitoring are required.

From “traditional” Deep Learning to Foundation Models

- **Traditional AI approach**

- Train task-specific neural networks

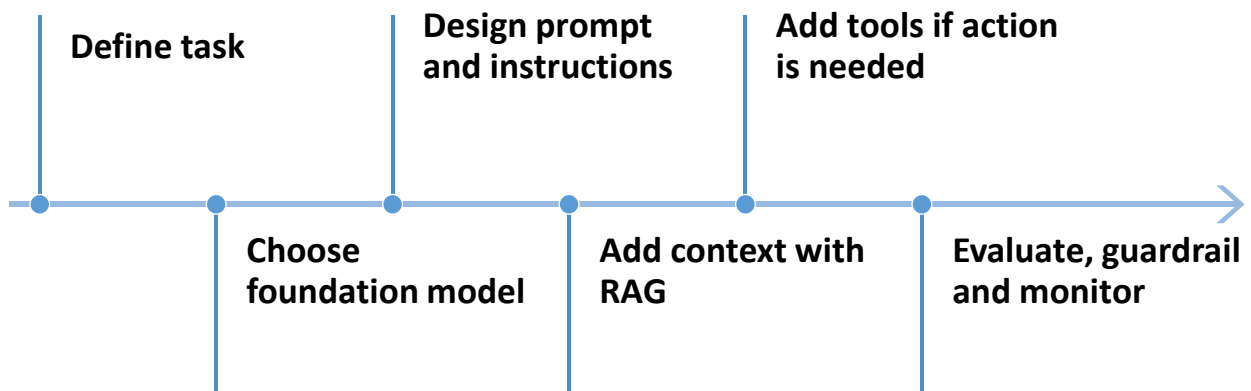
Dataset → model → prediction

- **Modern AI approach**

- Build around pre-trained foundation models

Prompt/context/tools → model → action

Modern AI application pipeline



The work shifts from “train one model” to “design and evaluate an AI system”.

Prompting, RAG, Fine-Tuning, Tools?

Need	Usually start with
Change tone, structure or behaviour	Prompt / system message
Use private or recent knowledge	RAG
Return a reliable data structure	Structured output / function schema
Call an external capability	Tool use / API call
Specialise behaviour at scale	Fine-tuning
Complete multi-step tasks	Agentic workflow

Evaluating AI agents

Intent

Did it understand the task?

Tool input

Were the arguments correct?

Planning

Did it choose sensible steps?

Grounding

Was the answer supported by evidence?

Tool choice

Did it call the right tool?

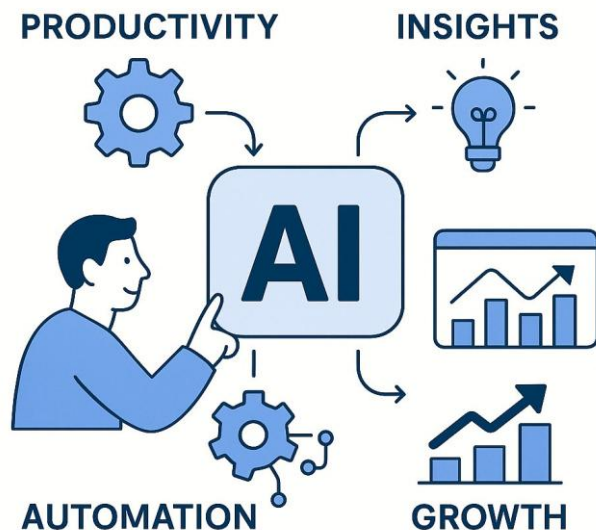
Safety

Should a human approve the action?

Opportunities of AI

- **Productivity and Automation**

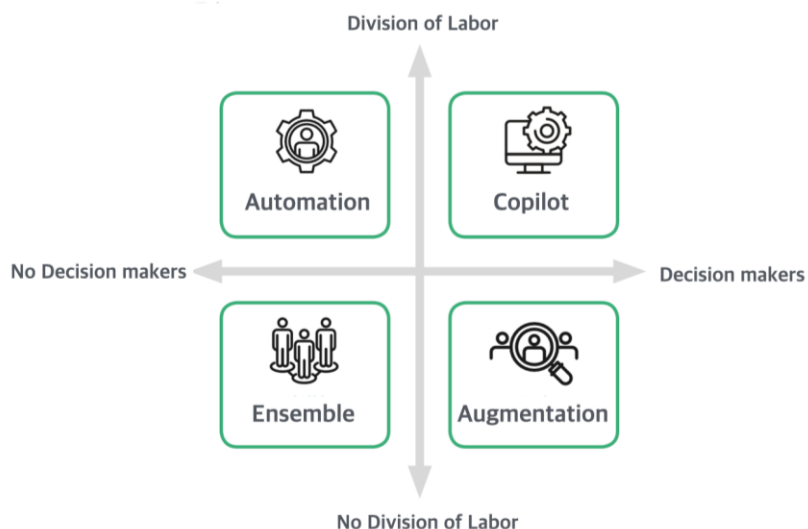
Generative AI tools streamline content creation, coding, and administrative tasks, offering time savings and scalability across sectors, from media to law to research.



Opportunities of AI

- **Human–AI Collaboration**

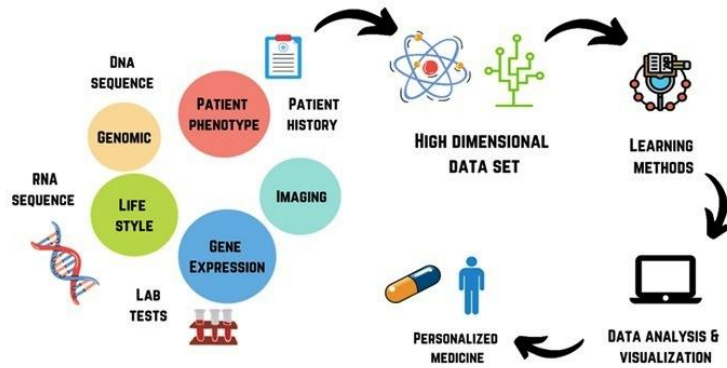
Agentic systems and AI copilots can act as creative or analytical partners, augmenting human decision-making in fields such as design, finance, and policymaking.



Opportunities of AI

• Personalised Healthcare and Education

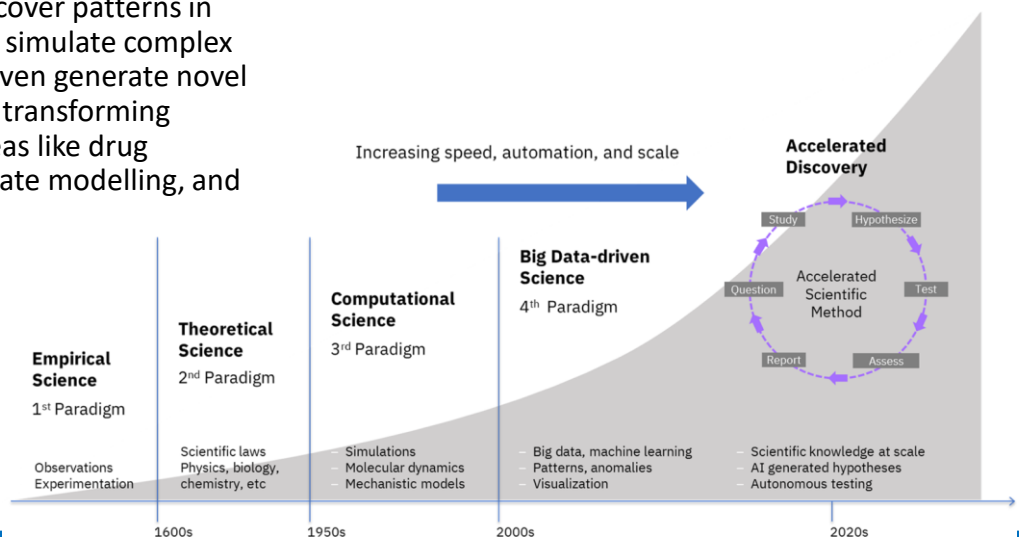
Through predictive modelling and adaptive systems, AI can tailor treatments to individual patients or learning pathways to students, improving outcomes and efficiency.



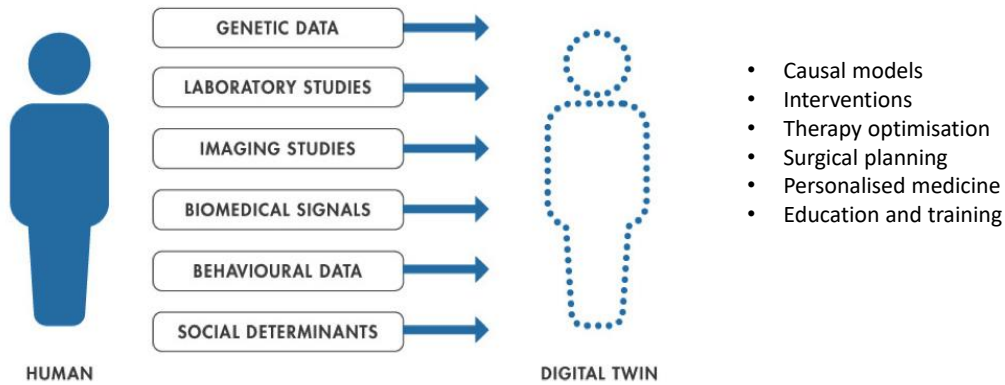
Opportunities of AI

• Accelerated Scientific Discovery

AI can help uncover patterns in large datasets, simulate complex systems, and even generate novel hypotheses — transforming research in areas like drug discovery, climate modelling, and genomics.



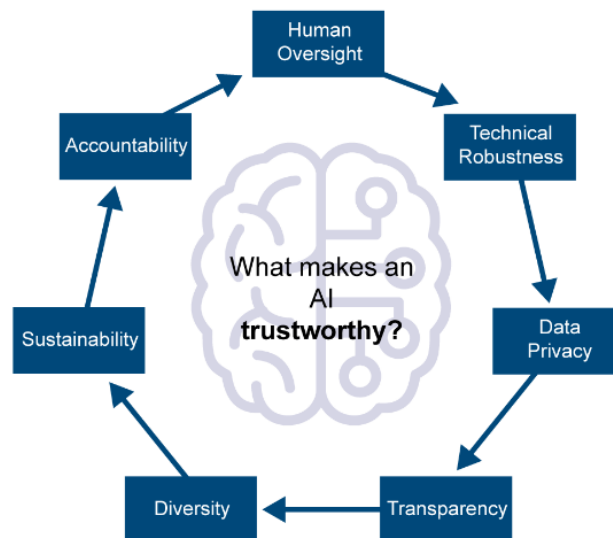
AI – Opportunities



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AI – Challenges

- Trustworthy AI



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Bias in AI

• Sources:

- Gender and racial bias, Linguistic bias,
- Socioeconomic bias, Political bias, etc

• Examples:

• Radiology and diagnostic imaging bias

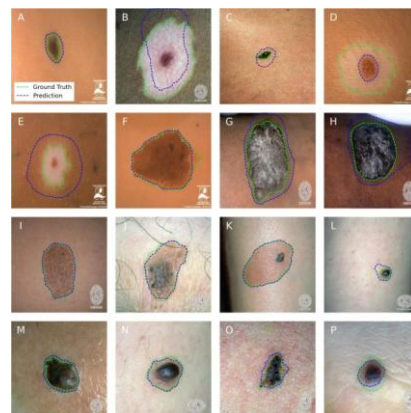
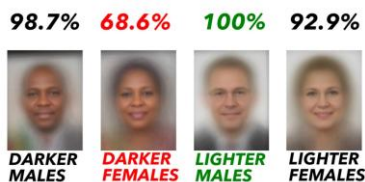
- AI trained on data from a single hospital or scanner type performs poorly on others.

• Under-representation in cardiology datasets

- Women and ethnic minorities are often **underrepresented** in cardiovascular AI training data.

• Skin-tone bias in dermatology imaging

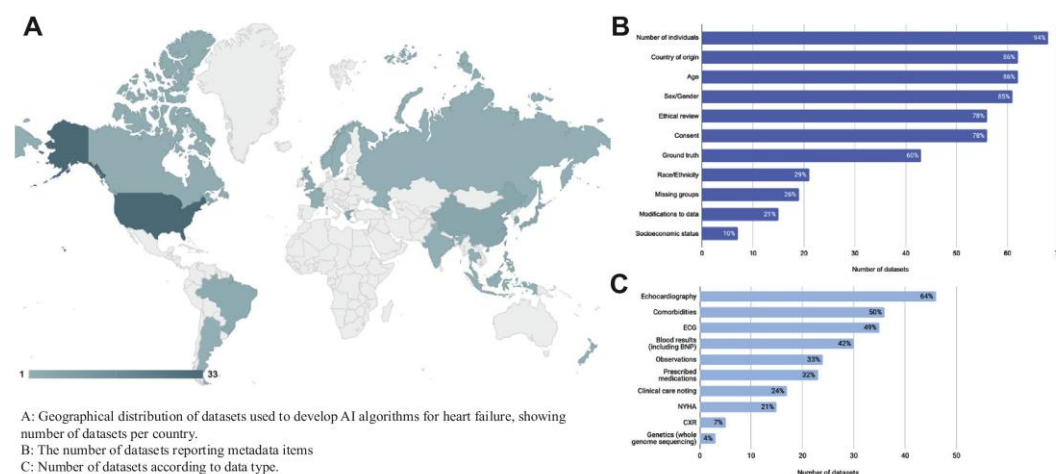
- AI skin lesion classifiers often underperform on darker skin tones due to training sets dominated by light-skinned images.
- Leads to delayed or incorrect melanoma detection.
(Example: 2022 study in *Lancet Digital Health* highlighting 60–70% performance gap.)



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CENTRAL ILLUSTRATION: Diversity and Inclusion Within Datasets in Heart Failure: A Systematic Review



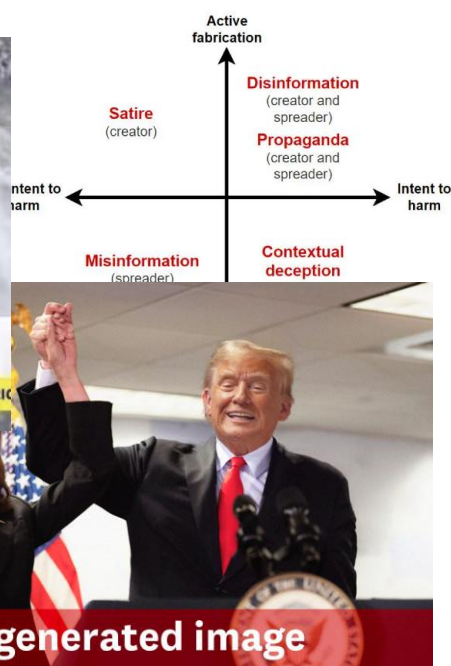
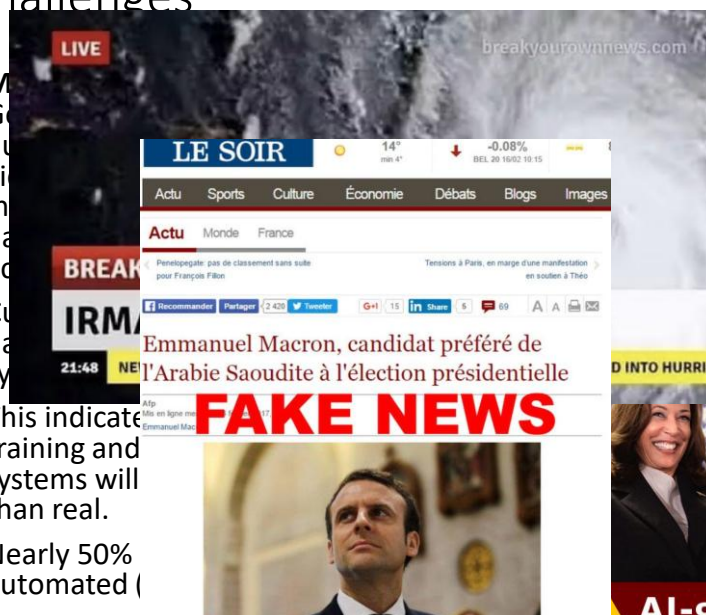
Laws E, et al. JACC Adv. 2025;4(3):101610.

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Challenges

- Misinformation and disinformation are becoming more prevalent in the digital age.
- This indicates that AI training and systems will be less than real.
- Nearly 50% automated (



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Challenges in AI – Risks of Diminishing Real Data for LLMs

1. Finite Supply of High-Quality Text

- Most high-quality human-written data (e.g. books, Wikipedia, web forums) has **already been used** to train current models like GPT-5 and Claude.
- Open-access web content is limited and increasingly protected by paywalls, copyright restrictions, or robots.txt exclusions.

2. Contamination by Synthetic Content

- As the web fills with **AI-generated text**, newer models risk training on **machine-produced data**, not human language.
- This can lead to **model degradation**, where LLMs learn from less diverse, repetitive, or subtly distorted patterns — a risk called **“model collapse”**.

3. Feedback Loops and Self-Amplification

- When LLMs train on outputs from earlier LLMs, subtle errors, hallucinations, and biases **amplify over generations**.
- This is analogous to **inbreeding in data**, reducing diversity and richness of language.

4. Legal and Ethical Barriers

- Laws like the EU AI Act and lawsuits around copyright (e.g., NYT vs OpenAI) are making it harder to obtain and use human-written content.

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AI Hallucinations



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S what is the world record for crossing the english channel entirely on foot

The world record for crossing the English Channel entirely on foot is held by Christof Wandratsch of Germany, who completed the crossing in 14 hours and 51 minutes on August 14, 2020. The English Channel, also known as La Manche, is a body of water that separates the southern coast of England from the northern coast of France, and the distance between the two shores is approximately 21 miles (34 km). Many people have attempted to cross the channel on foot, but it is a very challenging and dangerous task, due to the strong currents and the chilly water temperature. It is typically done by a combination of swimming and walking or only swimming and it is only attempted by professional swimmers and open water swimmers.

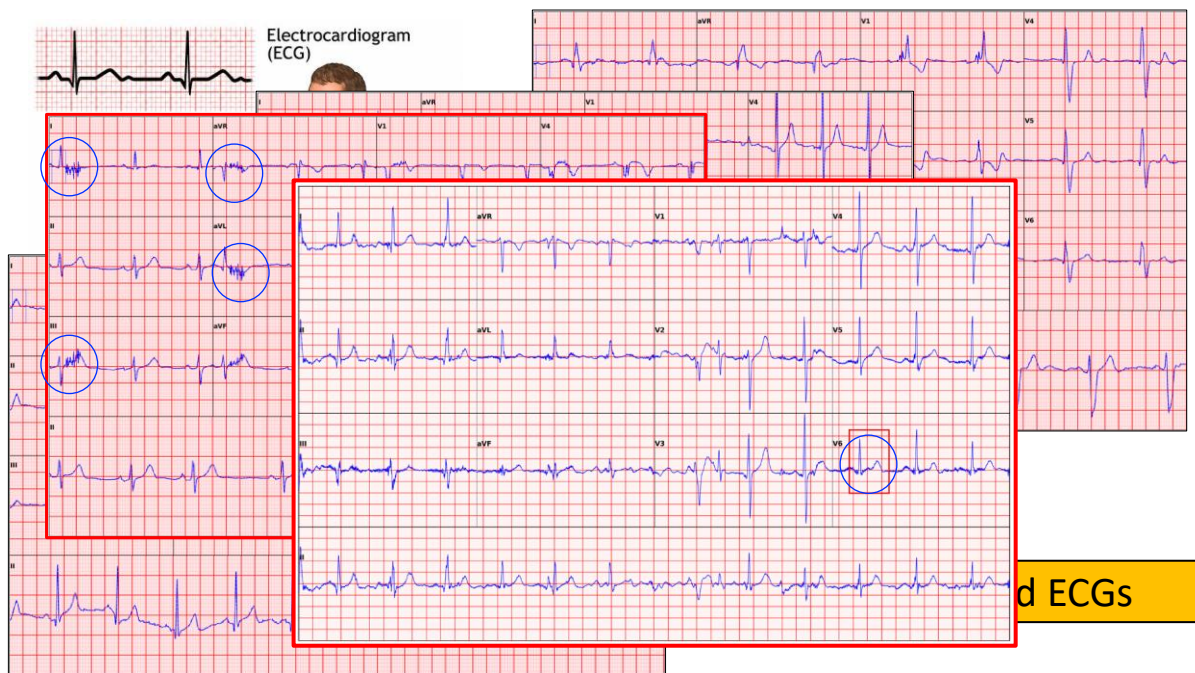
Regenerate response



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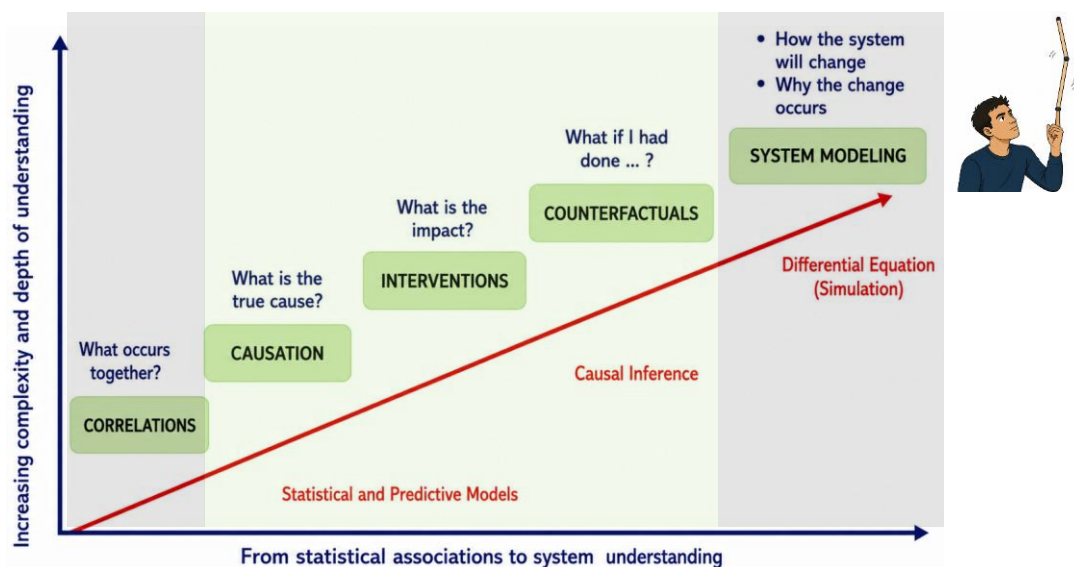
The issue with current AI ...

LLMs are fundamentally statistical predictive models:

- They learn patterns and *associations* in data.
- Specifically, they predict the next token given context

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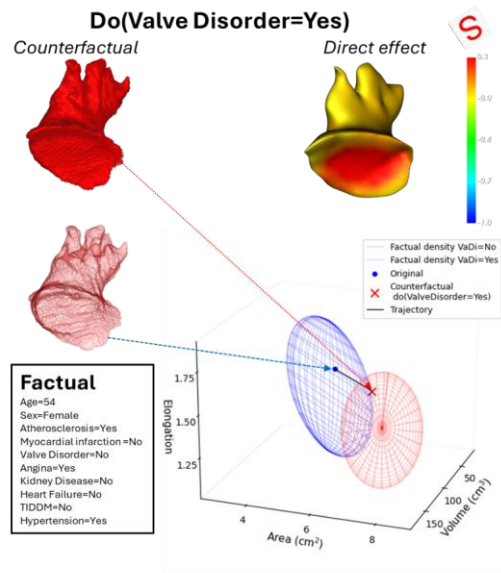
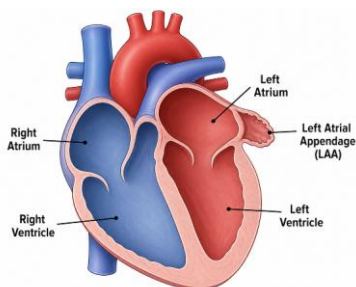
The case for *Causal AI*



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Causal modelling of AF-related stroke

- Atrial fibrillation (AF) is an abnormal and irregular heart rhythm affecting the atria
- In AF, abnormal blood flow in the left atrial appendage (LAA) can promote clot formation.
- If a clot travels to the brain, it can block blood flow and cause a stroke



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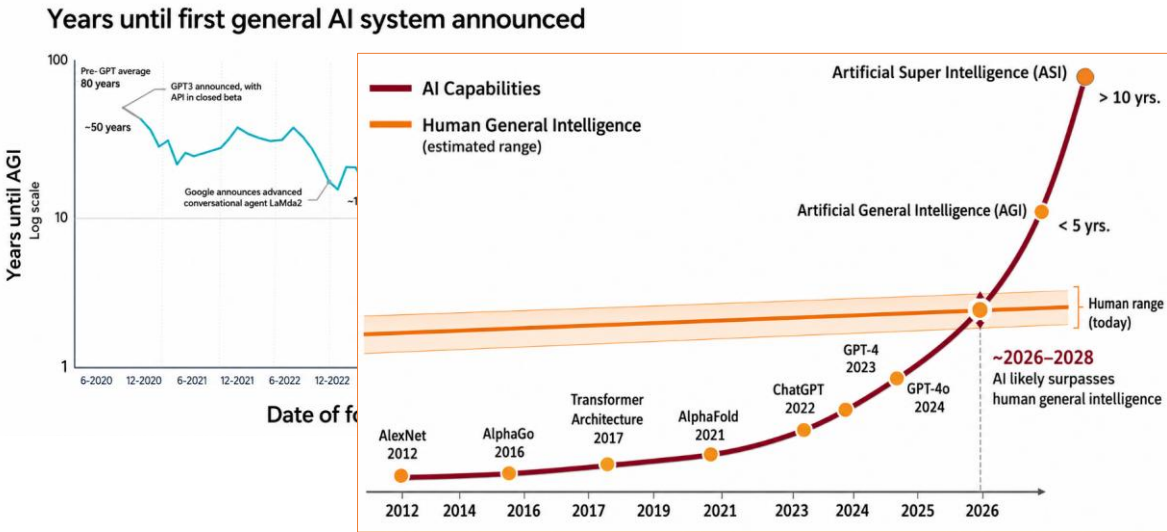
Causal AI underpins my current research

- **ARISTOTELES (EU Horizon Europe)**
 - Causal AI models for predicting risk trajectories of death, hospitalisation, and comorbidities.
 - Integration of multiple data modalities, including genetics, lifestyle, and clinical data.
- **TARGET (EU Horizon Europe)**
 - Causal AI models are central to the development of digital twins.
 - Development of hybrid causal digital twins of the heart, brain, and neuromusculoskeletal system, integrating mechanistic and data-driven models.
- **LIV.INNO (UKRI)**
 - Causal AI methods for scientific data generation.
 - Completion of astronomy surveys with information that has not yet been observed.
 - Development of a digital twin of the Milky Way.



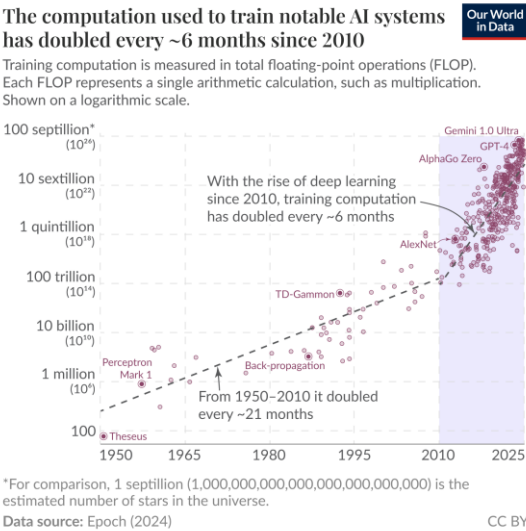
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The AI revolution



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Are we really getting there?



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Artificial General Stupidity

How ChatGPT and GPT-5 Are Overhyped BS and the AGI Singularity Is Cancelled

INNOVATION > AI

GPT-5 Is Launched But The AI Clearly Is Neither AGI Or Artificial Superintelligence

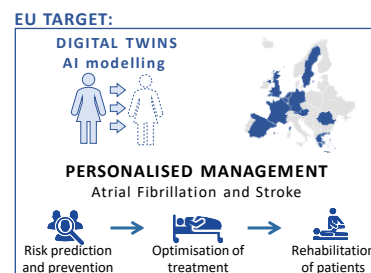
LLMs: A Dead End?

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Developing AI for healthcare



- We are currently leading TARGET, a €10m Horizon EU project on digital twins for the personalised management of stroke.
 - A digital twin, in healthcare, refers to the digital representation of individuals based on their specific characteristics, and medical and health status history.
 - TARGET's digital twins aim to improve patient outcomes by enabling more accurate diagnoses, personalised treatment plans, and efficient resource allocation.
- We are also part of another €6m Horizon EU project called ARISTOTELES.
 - In ARISTOTELES, we are leading the development of trustworthy AI solutions for the personalised prediction and early detection of comorbidities and multimorbidities.
 - This will help understand what factors (e.g., lifestyle choices) need to be changed to achieve a desired outcome (e.g., reducing the risk of diabetes).



Developing AI for freedom of expression



- Mapping the global free expression landscape worldwide.
 - We developed a novel freedom of expression index to gain a nuanced and data-led understanding of the level of censorship across the globe.
 - This was done in collaboration with Index on Censorship (IoC), an NGO that campaigns for and defends free expression worldwide.



Developing AI for astrophysics



- We co-host the Liverpool Centre for Doctoral Training for Innovation in Data Intensive Science (LIV.INNO)
 - We address the data challenges presented by research in astronomy, nuclear, theoretical and particle physics, accelerator science, mathematical and computer science.
- We use AI to investigate
 - How to reconstruct the history of our galaxy.
 - High energy physics.
 - The complex relationship between galaxies and their dark matter haloes.

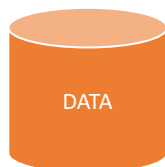


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From prediction to decision-making



Facts about
the world



Our current
interpretation
of the world

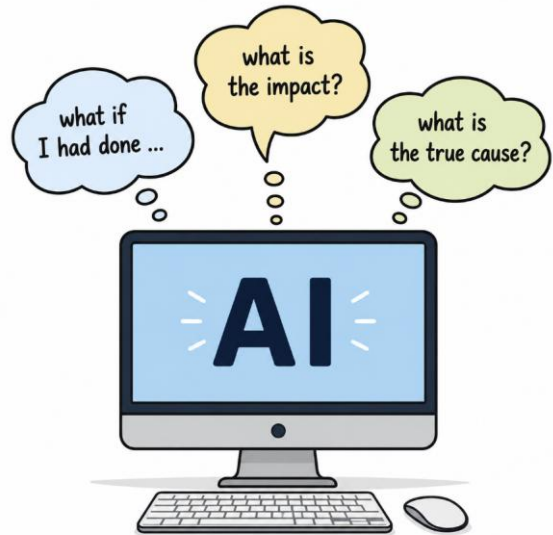
Prediction → Automation → Understanding → Decision-Making



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Closing reflections

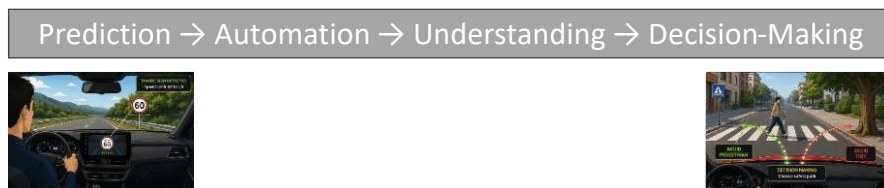
- Much of modern AI is still fundamentally based on prediction from data.
- Truly trustworthy AI may require models that capture structure, causality, and mechanisms.
- The challenge ahead is moving from prediction towards understanding.



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So ... prediction isn't enough

The future of AI is not only prediction.
... It is understanding.



THANK YOU.

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